Introduction to Econometrics
Week 2 Part 3: Finite sample properties of s^2 (section 1.3)

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Based on Rob Alessie lecture notes

Outline

- Finite sample properties of s^2
- Estimate of the variance $Var(\mathbf{b}|\mathbf{X})$
- Example (wage equation)

Finite sample properties of s^2

- We have defined before an estimator s^2 for the variance of the error term σ^2

$$s^2 = \frac{\mathbf{e}'\mathbf{e}}{n-K}$$

  where $n-K$ is called the **degrees of freedom**

Now, we want to

- prove that it is unbiased ($E(s^2|\mathbf{X}) = \sigma^2$)
- use it to get the estimator for the variance of OLS estimator ($\mathbf{b}$) $\Rightarrow$ we know that $Var(\mathbf{b}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ so once we have an ubiased estimator for σ^2, we can use it to estimate $Var(\mathbf{b}|\mathbf{X})$ and get:

$$\widehat{Var(\mathbf{b}|\mathbf{X})} = s^2(\mathbf{X}'\mathbf{X})^{-1} \tag{1}$$

Finite sample properties of s^2

Proposition 1.2 (unbiasedness of s^2): Under assumptions 1.1-1.4

$$E(s^2|\mathbf{X}) = \sigma^2$$

provided $n > K$
Proof:

- We should show that

$$E(\mathbf{e}'\mathbf{e}|\mathbf{X}) = (n-K)\sigma^2$$

  or equivalently (cf. Lemma 1 on next slide):

$$E(\varepsilon'\mathbf{M}\varepsilon|\mathbf{X}) = \sigma^2(n-K)$$

- The proof consists of two parts
  1. $E(\varepsilon'\mathbf{M}\varepsilon|\mathbf{X}) = \sigma^2 trace(\mathbf{M})$ (slide 7)
  2. $trace(\mathbf{M}) = n - K$ (slide 9)

Finite sample properties of s^2

First we need to lay some ground work...

- **Lemma 1**: $\mathbf{e}'\mathbf{e} = \varepsilon'\mathbf{M}\varepsilon$

Proof:
Note that:

$$\begin{aligned}
\mathbf{e} &= \mathbf{y} - \mathbf{X}\mathbf{b} \\
&= \mathbf{y} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \\
&= (\mathbf{I}_n - \mathbf{P})\mathbf{y} \\
&= \mathbf{M}\mathbf{y} \\
&= \mathbf{M}(\mathbf{X}\beta + \varepsilon) = \mathbf{M}\varepsilon
\end{aligned}$$

Since $\mathbf{M}$ is a symmetric and idempotent matrix it holds that

$$\mathbf{e}'\mathbf{e} = \varepsilon'\mathbf{M}'\mathbf{M}\varepsilon = \varepsilon'\mathbf{M}\varepsilon$$

q.e.d.

Finite sample properties of s^2

Some facts about the trace operator:

- **Fact 1:** The **trace** of a square matrix $\mathbf{A}$ is the sum of its diagonal elements of $\mathbf{A}$: $trace(\mathbf{A}) = \sum_i a_{ii}$
- **Fact 2:** The trace operator is linear $\Rightarrow$ $trace(\mathbf{A} + \mathbf{B}) = trace(\mathbf{A}) + trace(\mathbf{B})$
- **Fact 3**: $trace(\mathbf{AB}) = trace(\mathbf{BA})$

Finite sample properties of s^2

Proof of unbiasdness of s^2 part 1

- Notice that $\varepsilon'\mathbf{M}\varepsilon = \sum_{i=1}^{n}\sum_{j=1}^{n} m_{ij}\varepsilon_i\varepsilon_j$
- Also

$$
\begin{aligned}
E(\varepsilon'\mathbf{M}\varepsilon|\mathbf{X}) &= E\left(\sum_{i=1}^{n}\sum_{j=1}^{n} m_{ij}\varepsilon_i\varepsilon_j|\mathbf{X}\right) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} m_{ij}E(\varepsilon_i\varepsilon_j|\mathbf{X}) \\
&= \sum_{i=1}^{n} m_{ii}E(\varepsilon_i^2|\mathbf{X}) \\
&= \sigma^2\sum_{i=1}^{n} m_{ii} = \sigma^2 trace(\mathbf{M})
\end{aligned}
$$

Finite sample properties of s^2

- In the second step above, we use that m_{ij}'s are a function of $\mathbf{X}$ (linearity property of conditional expectations)
- In the third step we exploit assumption 1.4 $E(\varepsilon\varepsilon'|\mathbf{X}) = \sigma^2\mathbf{I}_n \Rightarrow E(\varepsilon_i\varepsilon_j|\mathbf{X}) = 0,$ for $i \neq j$

Finite sample properties of s^2

Proof of unbiasdness of s^2 part 2: $trace(\mathbf{M}) = n - K$

- We know that

$$\begin{aligned} \textbf{Projection matrix P} &= \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \\ \textbf{Annihilator M} &= \mathbf{I}_n - \mathbf{P} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \end{aligned}$$

- $$\begin{aligned} trace(\mathbf{P}) &= trace(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \\ &= trace((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}) \\ &= trace(\mathbf{I}_K) = K \end{aligned}$$

  In the second step we use the property

  $$trace(\mathbf{AB}) = trace(\mathbf{BA})$$

  ($\mathbf{A} = \mathbf{X}$ and $\mathbf{B} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$)

- $$\begin{aligned} trace(\mathbf{M}) &= trace(\mathbf{I}_n - \mathbf{P}) \\ &= trace(\mathbf{I}_n) - trace(\mathbf{P}) = n - K \end{aligned} \tag{2}$$

Finite sample properties of s^2

Intuition proposition 1.2

- Dividing $\mathbf{e}'\mathbf{e}$ by $n-K$ rather than by n makes this estimate unbiased for σ^2.
- The intuitive reason is that K parameters ($\boldsymbol{\beta}$) have to be estimated before obtaining the residual vector $\mathbf{e}$.
- More specifically, $\mathbf{e}$ has to satisfy the K normal equations, which limits the variability of the residual.

Once we have an estimator for σ^2, we can get an estimator for the variance of b ($Var(\mathbf{b}|\mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$):

$$\widehat{Var(\mathbf{b}|\mathbf{X})} = s^2(\mathbf{X}'\mathbf{X})^{-1} \tag{3}$$

Example: wage and education

We consider the following simple regression model

$$wage_i = \beta_1 + \beta_2 educ_i + \varepsilon_i \tag{4}$$

where

$wage_i$=wage is measured in dollars per hour

$educ_i$=years of schooling

- Model (3) has been estimated using Stata

Example: wage and education

```
    . sum wage,detail
    average hourly earnings
-------------------------------------------------------------
    Percentiles Smallest
 1%       1.67       .53
 5%       2.75      1.43
10%       2.92       1.5  Obs                 526
25%       3.33       1.5  Sum of Wgt.         526

50%       4.65            Mean           5.896103
                 Largest  Std. Dev.      3.693086
75%       6.88     21.86
90%         10      22.2  Variance       13.63888
95%         13     22.86  Skewness       2.007325
99%         20     24.98  Kurtosis       7.970083
```

Notice that the distribution of *wage* is rightly skewed and has fat tails (cf. skewness and kurtosis)

Example: wage and education

Regression results

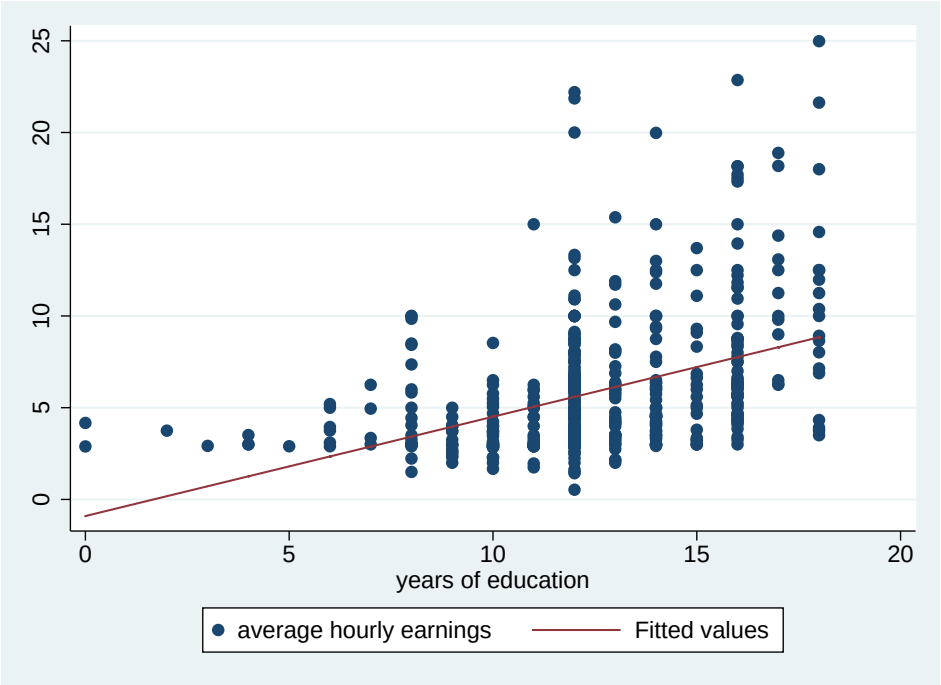
```
. reg wage educ
  Source |       SS       df       MS              Number of obs =     526
---------+------------------------------           F(1, 524)=103.36
   Model |1179.73204     1  1179.73204           Prob> F =0.0000
Residual |5980.68225   524  11.4135158           R^2     =0.1648
---------+------------------------------           Adj R^2 =0.1632
   Total |7160.41429   525  13.6388844           Root MSE=3.3784

------------------------------------------
    wage |      Coef.   Std. Err.      t
---------+--------------------------------
    educ |   .5413593    .053248    10.17
   _cons |  -.9048516   .6849678    -1.32
------------------------------------------
```

Example: wage and education

- b_{educ}=.541
  Interpretation: One year of extra schooling leads to 54 dollar cents extra wage per hour.
- $s^2 = e'e/(n-K)$=(5980.68/524=11.41)
- **Standard error of the regression**: $s = \sqrt{s^2} = \sqrt{11.41}$= 3.3784 (cf. "*rootMSE*")
- $se(b_{educ}|X) = \sqrt{\widehat{Var}(b_{educ}|X)} = .053$
- $R^2 = 1 - \frac{e'e}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \frac{5980.68}{7160.41} = \frac{1179.73}{7160.41} = 0.165$
- Adjusted $R^2 = 1 - \frac{e'e/(n-K)}{(\sum_{i=1}^{n}(y_i - \bar{y})^2)/(n-1)} = 1 - \frac{11.414}{13.639}$=0.1632

Example: wage and education



- The figure shows evidence of heteroskedasticity: the absolute value of the residuals increase with *educ*: standard errors should be viewed with care
- Simple solution: consider the following semi-log specification:

$$\log(wage_i) = \beta_1 + \beta_2 educ_i + \varepsilon_i \tag{5}$$

- Taking the log makes the distribution of the wage less rightly skewed (check in the data)

Example: wage and education

Regression results

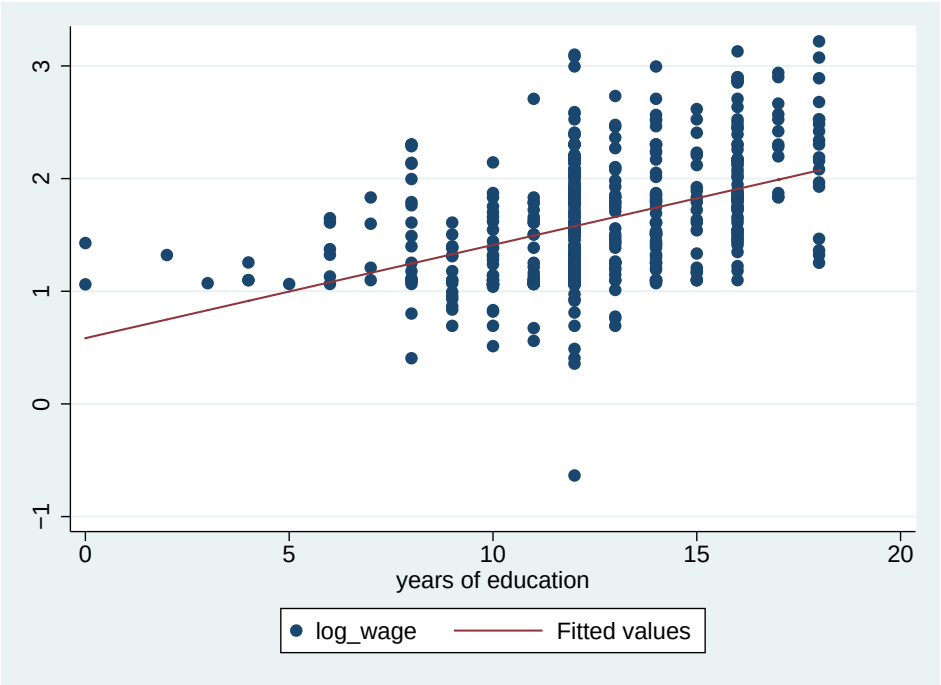
```
. reg log_wage educ

  Source |       SS       df       MS              Number of obs =     526
---------+------------------------------           F(  1,   524) =  119.58
   Model |27.5606288     1  27.5606288           Prob > F      =  0.0000
Residual |120.769123   524  .230475425           R-squared     =  0.1858
---------+------------------------------           Adj R-squared =  0.1843
   Total |148.329751   525   .28253286           Root MSE      =  .48008

------------------------------------------
log_wage |      Coef.   Std. Err.      t
---------+--------------------------------
    educ |   .0827444   .0075667    10.94
   _cons |   .5837727   .0973358     6.00
------------------------------------------
```

Interpretation b_{educ}: The return to education is equal to 8.27 percent (due to a one year increase in the *educ* (years of schooling), the hourly wage rate increases with 8.27 percent (=.0827444*100)

Example: wage and education



The absolute value of the residuals is no longer increasing with the level of education

Interpretation of the coefficients in different model specifications

Table: Summary of functional forms involving logarithms

Model	Dependent Variable	Independent Variable	Interpretation of β
level-level	y	x	$\Delta y = \beta \Delta x$
level-log	y	$\ln(x)$	$\Delta y \approx \frac{\beta}{100} \% \Delta x$
log-level (semi-log)	$\ln(y)$	x	$\% \Delta y \approx 100 \beta \Delta x$[1]
log-log	$\ln(y)$	$\ln(x)$	$\% \Delta y = \beta \% \Delta x$

1: The 'log-level' approximation is precise for small values of β.

1

A note on the semi-log specification

- A useful approximation for the natural logarithm for small x is

$$\ln(1+x) \approx x$$

  This linear approximation x is very close for $|x| < 0.1$, and reasonably close for $0.1 < |x| < 0.2$, but the difference increases with $|x|$.
- Now, if y^* is $c\%$ greater than y, then

$$y^* = (1 + c/100)y.$$

  Taking natural logarithms,

$$\ln(y^*) = \ln(y) + \ln(1 + c/100)$$

  or

$$\ln(y^*) - \ln(y) = \ln(1 + c/100) \approx \frac{c}{100} = \beta$$

  or $c \approx \beta * 100$ (see table 1)
- Exact calculation of the percentage difference

$$100\frac{y - y^*}{y^*} = 100(\exp(\ln(y^*) - \ln(y)) - 1) = 100(e^{\beta} - 1)$$

  you should use this formula if β is "large" (rule of thumb $|\beta| > 0.2$).